

APQP AND PPAP RELATIONSHIP

How APQP Phase 4 Outputs become PPAP 18 Elements | Visual Guide | Quality Head Agents

THE RELATIONSHIP

APQP and PPAP are two of the Five AIAG Core Tools that are closely linked. PPAP is the output of APQP — specifically Phase 4 (Product and Process Validation). The PPAP package demonstrates that the APQP process was completed successfully.

SIMPLE EXPLANATION

Think of it this way:

- APQP = The process (journey from customer requirement to production)
- PPAP = The proof (evidence that the journey was completed correctly)
- PSW = The final sign-off (customer confirmation that PPAP is accepted)

APQP PHASE → PPAP ELEMENT MAPPING

APQP PHASE	APQP DELIVERABLE	PPAP ELEMENT #	PPAP ELEMENT NAME
Phase 1	Design Goals, VOC	—	Background for design requirements
Phase 2	DFMEA	3	Design Failure Mode and Effect Analysis
Phase 2	Engineering Drawings	1	Design Records
Phase 2	Engineering Change Documents	2	Authorized Engineering Change Documents
Phase 2	Material Specifications	9	Material Performance Test Results
Phase 2	Design Verification (DVP)	11	Design Verification
Phase 2	Design Review Summary	17	Customer Specific Requirements
Phase 3	Process Flow Diagram (PFD)	4	Process Flow Diagram
Phase 3	PFMEA	5	Process Failure Mode and Effect Analysis
Phase 3	Control Plan (Pre-Launch)	6	Control Plan (final version)
Phase 3	Measurement System Analysis Plan	15	Measurement System Analysis Studies
Phase 3	Packaging Specifications	18 (via PSW)	Part Submission Warrant — packaging
Phase 4	Production Trial Run (PTR)	10	Production Part Approval
Phase 4	Dimensional Results	7	Measurement System Analysis Studies
Phase 4	Material/Performance Test Results	8	Dimensional Results
Phase 4	MSA / GRR Studies	15	Measurement System Analysis Studies
Phase 4	Capability Study (Cpk)	16	Qualified Laboratory Documentation
Phase 4	Appearance Approval (AAR)	12	Appearance Approval Report
Phase 4	Sample Parts	13	Sample Production Parts
Phase 4	Master Sample	14	Master Sample
Phase 4	Checking Aids	16	Checking Aids
Phase 4	Production Control Plan (Final)	6	Control Plan — Production version

APQP PHASE	APQP DELIVERABLE	PPAP ELEMENT #	PPAP ELEMENT NAME
Phase 4	PSW Sign-off	18	Part Submission Warrant (PSW)

PPAP SUBMISSION LEVELS

LEVEL	DESCRIPTION	WHAT IS SUBMITTED TO CUSTOMER	TYPICAL USE CASE
Level 1	Warrant only, and where required, approved AAR only	PSW and AAR only	Minor changes, customer specific direction
Level 2	Warrant with product samples and limited supporting data	PSW + samples + selected documents	Minor new parts, low risk
Level 3	Warrant with product samples and complete supporting data	PSW + samples + all 18 elements	Standard new part submission (most common)
Level 4	Warrant and other requirements as specified by customer	As specified by customer	Defined in customer-specific requirements
Level 5	Warrant with product samples and complete supporting data	Complete supporting data + PSW + Part supplier participation	High-risk or critical part or new suppliers

WHEN IS PPAP REQUIRED?

- New part or product designed by the supplier
- Correction of a discrepancy on a previously submitted part
- Product modified by an engineering change to design records, specifications, or materials
- Use of another optional construction or material
- Production from new or modified tools (except perishable tools), dies, molds, patterns, etc.
- Production following refurbishment or rearrangement of existing tooling or equipment
- Production transferred to a different plant location or from an additional plant location
- Change of subcontractor for parts, non-equivalent materials, or services (e.g., heat treating, plating)
- New source of raw material from new or existing subcontractor
- Change in test/inspection method — new technique, no design record change
- After production has been discontinued for 12 months or more
- After any change to the process, equipment, or environment that may affect form, fit, or function